

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Mathematics — Class IX

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 1

General Instructions:

1. Attempt all questions from Section A (MCQs). Bubble the correct answer in the Answer column.
2. Attempt all 11 questions from Section B. For each question, attempt either the main part **OR** its alternative — not both.
3. Attempt all 4 questions from Section C. For each question, attempt either the main part **OR** its alternative — not both.
4. Write legibly. Show all working clearly.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	$a(b + c - d)$ equals:	$a(b + c + d)$	$ab + ac - ad$	$ab + ac + ad$	$ab - ac - ad$	○○○○
2	Which number is its own multiplicative inverse (other than 1)?	3	-3	-1	0	○○○○
3	If $\log(x + 3) = \log(15x - 4)$, then x equals:	0.5	7	14	2	○○○○
4	The characteristic of $\log 0.00327$ is:	2	-3	3	0	○○○○
5	$x(2a + 3b + 7) + x(2a + 3b + 5)$ equals:	$2x(2a + 3b + 6)$	$4x(2a + 3b + 6)$	$x(2a+3b+7)(2a+3b+5)$	$(x+x)(2a+3b+12)$	○○○○
6	The sum of $2a/(a^2-1)$ and $-a/(a^2-1)$ is:	$3a/(a^2-1)$	$a/(a^2-1)$	$(2a-a)/[2(a^2-1)]$	$-2a/(a^2-1)$	○○○○
7	The square root of $36x^6y^{16}$ is:	$6x^6y^{16}$	$6xy$	$6x^3y^8$	$16x^3y^8$	○○○○
8	What number must be subtracted from the right side of $7x = 30$ so that 4 is a solution?	7	2	4	-2	○○○○
9	The solution set of $ x + 7 = 3$ is:	$\{-4\}$	$\{4, -4\}$	$\{ \}$ (empty)	$\{-7, -3\}$	○○○○
10	If the distance between $A(c, 10)$ and $B(0, -1)$ is 20 units, the value of c is:	279	$3\sqrt{31}$	3	40	○○○○
11	The midpoint of the hypotenuse of the right triangle with vertices $(4,0)$, $(2,1)$, $(-1,-5)$ is:	$(2, 1/2)$	$(1/2, -2)$	$(3/2, -5/2)$	$(3/2, 5/2)$	○○○○
12	If the endpoints of a diameter of a circle are $(4,5)$ and $(6,9)$, the centre is:	$(7, 5)$	$(1, 2)$	$(5, 7)$	$(0, 0)$	○○○○

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. Answer the main part OR its alternative for each.

Part	Question	Mks	OR	Alternative Question	Mks
2(i)	Simplify using laws of exponents: $[(-2)^3 \cdot (-2)^{-4} \cdot (-2)] \div (-2)^{-3}$	3	OR	Write as an equivalent radical/simplify: (a) $(-27)^{1/3}$ (b) $(\sqrt{16})^4$ (c) $(\sqrt[3]{-8})^9$	3
2(ii)	Identify the property illustrated in each: (a) $4 \times \frac{1}{4} = 1$ (b) $6 + 0 = 6$ (c) $2 \times (3 \times 5) = (2 \times 3) \times 5$	3	OR	State TRUE or FALSE (give a reason/counterexample if false): (a) Every real number is an irrational number. (b) Every real number is either rational or irrational.	3
2(iii)	Using laws of logarithms: (a) Expand $\log(5pq^2/xy^3)$ (b) Combine into a single log: $3\log x - 5\log y$	3	OR	Evaluate: (a) $\log_9(\sqrt[3]{9})$ (b) $\log_{\sqrt{7}}(343)$	3
2(iv)	If $\log_b 2 = 0.3010$, $\log_b 3 = 0.4771$, $\log_b 5 = 0.6990$, evaluate: (a) $\log_b(6/5)$ (b) $\log_b(\sqrt[3]{450}/\sqrt{27})$	3	OR	Evaluate $\log_b(0.024)$ using the same given values. Show full working.	3
2(v)	Factorize completely: (a) $x^4 - 12x^2 + 16$ (b) $2m^4 + m^2n^2 - 3n^4$	3	OR	Factorize by substitution: $(y^2 + 2y - 3)(y^2 + 2y + 11) + 48$	3
2(vi)	The expression $a^6 + 729$ can be written as a sum of two squares or a sum of two cubes. Which form allows factorization, and why? Factorize it.	3	OR	Express $8 + 12t + 6t^2 + t^3$ as a product of three factors. State whether each factor is a binomial or a trinomial.	3
2(vii)	Factorize by substitution: $(2x^2 - 3x + 6)(2x^2 - 3x) - 55$	3	OR	Find the HCF of $x^3 + 3x^2 - 8x - 24$ and $x^3 + 3x^2 - 3x - 9$ by factorization.	3
2(viii)	Solve for x: $(2x-11)/12 = (2x+10)/12 - [(28-2x)/4 - 1/4]$	3	OR	Solve: $5 - 5y + 1 = -9$	3
2(ix)	Solve the compound inequality: $(3/2)x \leq -3$ or $(2/3)x \geq 4$, $x \in \mathbb{R}$	3	OR	Solve: $4(2y+3) - (6y-1) > 10$, $y \in \mathbb{R}$	3
2(x)	By the midpoint formula, prove the diagonals of rhombus A(-3,-1), B(0,0), C(1,3), D(-2,2) bisect each other. Also find the length of each diagonal.	3	OR	A(3,4), B(-1,7) and C(x,y) are collinear points such that B is the midpoint of AC. Find x and y.	3
2(xi)	Show that A(0,0), B(0,5), C(4,7), D(4,2) are the vertices of a parallelogram (use the midpoint-of-diagonals method).	3	OR	P(2,3) is the midpoint of the segment joining A(x,4) and B(-3,y). Find x and y.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt all 4 questions. Answer the main part OR its alternative for each.

Part	Question	Mks	OR	Alternative Question	Mks
3(i)	Given $\log_b 2 = 0.3010$, $\log_b 3 = 0.4771$, $\log_b 5 = 0.6990$, evaluate $\log_b 0.0036$ using laws of logarithms. Show full working, including the adjustment needed to make the mantissa positive.	5	OR	Evaluate using logarithm tables: $\sqrt[3]{129.4} \div \sqrt[7]{27.37}$ (7th root of 129.4 divided by cube root of 27.37)	5
3(ii)	Factorize completely: $x^3p^2 - 8y^3p^2 - 4x^3q^2 + 32y^3q^2$	5	OR	Simplify (all denom. non-zero): $[(x^2+x-2)/(x^2-x-20)] \times [(x^2+5x+4)/(x^2-x)] \div \{ [(x^2+3x+2)/(x^2-2x-15)] \times [(x+3)/x^2] \}$	5
3(iii)	(a) Solve: $\sqrt{a - 1/2} = \sqrt{2a/5 + 2/5}$ [3] (b) The difference between three times a number y and 18 is less than 12 or greater than 39. What real numbers does y represent? [2]	5	OR	(a) Solve: $[\sqrt{5x-4} - 4]/10 = -1$. State clearly if the solution is extraneous. [3] (b) Solve: $1.3x - 0.2 = 0.3x - 1.5$ [2]	5
3(iv)	Show that A(2,-1), B(8,-1), C(8,3), D(2,3) are the vertices of a rectangle. Hence show that triangles ABC and ABD (formed by diagonals AC and BD) are both right triangles.	5	OR	Triangle ABC has vertices A(3,7), B(-4,0), C(4,0). D and E are the midpoints of AB and AC. (a) Find D and E (b) Find d(D,E) and d(B,C) (c) Compare the lengths and state the geometric relationship.	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Unit 1 (Real Numbers) | Unit 2 (Logarithms) | Unit 4 (Factorization & Algebraic Manipulation) | Unit 5 (Linear Equations & Inequalities) | Unit 7 (Coordinate Geometry)

— End of Question Paper —